

Numericals based on De Broglie Hypothesis

1. An electron beam is accelerated from rest through a potential difference of 200 V.

i) Calculate the associated wavelength.

ii) This beam is passed through a diffraction grating of spacing $3 \text{ \AA}$. At what angle of deviating from the incident direction will be the first maximum observed?

i) Given: $h = 6.63 \times 10^{-34} \text{ J.s}$, $m = 9.1 \times 10^{-31} \text{ kg}$, $e = 1.6 \times 10^{-19} \text{ C}$

$V = 200 \text{ volts}$, $\lambda = ?$

$$\lambda = \frac{h}{\sqrt{2meV}} = \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-19} \times 200}}$$
$$= 8.688 \times 10^{-11} \text{ m} = 0.8688 \text{ \AA}$$

ii) $d = 3 \times 10^{-10} \text{ m}$, order $m = 1$

$$2d \sin \theta = n\lambda$$

$$\therefore \theta = \sin^{-1} \left(\frac{\lambda}{2d} \right) = \sin^{-1} \left(\frac{8.688 \times 10^{-11}}{2 \times 3 \times 10^{-10}} \right) = 8.32^\circ$$

2. Calculate the wavelength of the wave associated with a neutron moving with energy 0.025 eV . Mass of neutron is $1.676 \times 10^{-27} \text{ kg}$.

Given: $h = 6.63 \times 10^{-34} \text{ J.s}$, $m = 1.676 \times 10^{-27} \text{ kg}$

$E = 0.025 \text{ eV} = 0.025 \times 1.6 \times 10^{-19} \text{ J}$

$\lambda = ?$

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}} = \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 1.676 \times 10^{-27} \times 0.025 \times 1.6 \times 10^{-19}}}$$
$$= 1.811 \times 10^{-10} \text{ m}$$

3. Calculate the de Broglie wavelength of the proton moving with a velocity equal to $\frac{1}{20}$ th of velocity of light. Mass of proton is 1.6×10^{-27} kg.

$$\text{Given : } h = 6.63 \times 10^{-34} \text{ J.s, } m = 1.6 \times 10^{-27} \text{ kg}$$

$$v = \frac{1}{20} \times 3 \times 10^8, \lambda = ?$$

$$\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{6.63 \times 10^{-34}}{1.6 \times 10^{-27} \times \frac{1}{20} \times 3 \times 10^8}$$

$$= 2.763 \times 10^{-14} \text{ m}$$

4. Calculate velocity and de Broglie wavelength of an α - particle of energy 1 keV. Mass of α - particle is 6.68×10^{-27} kg.

$$\text{Given : } h = 6.63 \times 10^{-34} \text{ J.s, } m = 6.68 \times 10^{-27} \text{ kg}$$

$$E = 1 \text{ keV} = 1000 \times 1.6 \times 10^{-19} \text{ J}$$

$$v = ? \quad \lambda = ?$$

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}} = \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 6.68 \times 10^{-27} \times 1000 \times 1.6 \times 10^{-19}}}$$

$$= 4.535 \times 10^{-13} \text{ m}$$

5. Calculate de Broglie wavelength associated with an α - particle accelerated by a potential difference of 100 kV. Mass of α - particle is 6.68×10^{-27} kg.

$$\text{Given : } h = 6.63 \times 10^{-34} \text{ J.s, } m = 6.68 \times 10^{-27} \text{ kg,}$$

$$q = 2e = 2 \times 1.6 \times 10^{-19} \text{ C, } V = 100 \times 10^3 \text{ volts, } \lambda = ?$$

$$\lambda = \frac{h}{\sqrt{2mqV}} = \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 6.68 \times 10^{-27} \times 2 \times 1.6 \times 10^{-19} \times 100 \times 10^3}}$$

$$= 3.206 \times 10^{-14} \text{ m}$$

6. Find the energy of neutron in units of electron volt whose de Broglie wavelength is $1 \text{ \AA}$. Given - mass of neutron $= 1.674 \times 10^{-27} \text{ kg}$

$$\text{Given : } h = 6.63 \times 10^{-34} \text{ J.s, } m = 1.674 \times 10^{-27} \text{ kg}$$

$$\lambda = 1 \text{ \AA} = 10^{-10} \text{ m, } E = ?$$

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}} \quad \therefore E = \frac{h^2}{2m\lambda^2} = \frac{(6.63 \times 10^{-34})^2}{2 \times 1.674 \times 10^{-27} \times (10^{-10})^2}$$

$$= 1.313 \times 10^{-20} \text{ J} = \frac{1.313 \times 10^{-20}}{1.6 \times 10^{-19}} = 0.082 \text{ eV}$$

7. A bullet of mass 40 gm and an electron both travel with the velocity of 1100 m/s. What wavelengths can be associated with them? Why the wave nature of bullet can not be revealed using diffraction effect?

i) For electron,

$$\text{Given: } h = 6.63 \times 10^{-34} \text{ J.s, } m = 9.1 \times 10^{-31} \text{ kg, } e = 1.6 \times 10^{-19} \text{ C}$$

$$v = 1100 \frac{\text{m}}{\text{s}}, \quad \lambda = ?$$

$$\begin{aligned} \lambda &= \frac{h}{mv} = \frac{6.63 \times 10^{-34}}{9.1 \times 10^{-31} \times 1100} \\ &= 6.623 \times 10^{-7} \text{ m} = 6623 \text{ \AA} \end{aligned}$$

ii) For bullet, Given: $h = 6.63 \times 10^{-34} \text{ J.s, } m = 40 \text{ gm} = 40 \times 10^{-3} \text{ kg,}$

$$v = 1100 \frac{\text{m}}{\text{s}}, \quad \lambda = ?$$

$$\begin{aligned} \lambda &= \frac{h}{mv} = \frac{6.63 \times 10^{-34}}{40 \times 10^{-3} \times 1100} \\ &= 1.507 \times 10^{-35} \text{ m} \end{aligned}$$

As the wavelength associated with bullet is of the order of 10^{-35} m , to reveal the wave nature of wave associated with bullet, a diffraction grating having width of the slit of the order of 10^{-35} m is needed. Such diffraction grating is not available. So the wave nature of the bullet can not be revealed.

8. Find the energy of neutron in units of electron volt whose de Broglie wavelength is $1 \text{ \AA}$. Given – mass of neutron = $1.674 \times 10^{-27} \text{ kg}$

$$\text{Given: } h = 6.63 \times 10^{-34} \text{ J.s, } m = 1.674 \times 10^{-27} \text{ kg}$$

$$\lambda = 1 \text{ \AA} = 10^{-10} \text{ m, } E = ?$$

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}}$$

$$\begin{aligned} E &= \frac{h^2}{2m\lambda^2} = \frac{(6.63 \times 10^{-34})^2}{2 \times 1.674 \times 10^{-27} \times (10^{-10})^2} \\ &= 1.313 \times 10^{-20} \text{ J} = \frac{1.313 \times 10^{-20}}{1.6 \times 10^{-19}} = 0.082 \text{ eV} \end{aligned}$$

9. An electron accelerated through 1000 volts and is reflected from a crystal. The first order reflection occurs when glancing angle is 70° . Calculate the interplanar spacing of the crystal.

$$\text{Given: } h = 6.63 \times 10^{-34} \text{ J.s, } m = 9.1 \times 10^{-31} \text{ kg,}$$

$$e = 1.6 \times 10^{-19} \text{ C, } V = 1000 \text{ volts, } n = 1, \quad \lambda = ? \quad d = ? ,$$

$$\lambda = \frac{h}{\sqrt{2meV}} = \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-19} \times 1000}}$$

$$= 3.885 \times 10^{-11} \text{ m}$$

Using Bragg's law,

$$2 d \sin \theta = n \lambda$$

$$\begin{aligned} \therefore d &= \frac{n \lambda}{2 \sin \theta} = \frac{1 \times 3.885 \times 10^{-11}}{2 \times \sin 70} \\ &= 2.067 \times 10^{-11} \text{ m} = 0.2067 \text{ \AA} \end{aligned}$$

10. An electrons accelerated through 100 volts are reflected from a crystal. Calculate the glancing angle at which the first order reflection occurs. Given lattice spacing = $2.15 \text{ \AA}$.

$$\text{Given: } h = 6.63 \times 10^{-34} \text{ J.s, } m = 9.1 \times 10^{-31} \text{ kg, } d = 2.15 \times 10^{-10} \text{ m}$$

$$e = 1.6 \times 10^{-19} \text{ C, } V = 100 \text{ volts, } n = 1, \lambda = ? \quad \theta = ?$$

$$\begin{aligned} \lambda &= \frac{h}{\sqrt{2meV}} = \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-19} \times 100}} \\ &= 1.229 \times 10^{-10} \text{ m} \end{aligned}$$

Using Bragg's law,

$$2 d \sin \theta = n \lambda$$

$$\begin{aligned} \therefore \theta &= \sin^{-1} \left(\frac{n \lambda}{2 d} \right) = \sin^{-1} \left(\frac{1 \times 1.229 \times 10^{-10}}{2 \times 2.15 \times 10^{-10}} \right) \\ \therefore \theta &= 16.61^\circ \end{aligned}$$

11. An electron and photon each have a wavelength of $2 \text{ \AA}$. What are their momentum and energy?

$$\text{Mass of neutron} = 1.676 \times 10^{-27} \text{ kg, } \text{Mass of electron} = 9.1 \times 10^{-31} \text{ kg,}$$

$$\text{Planck's constant} = 6.63 \times 10^{-34} \text{ J.s}$$

Given:

$$\text{Mass of neutron} = 1.676 \times 10^{-27} \text{ kg, } \text{Mass of electron} = 9.1 \times 10^{-31} \text{ kg,}$$

$$\text{Planck's constant} = 6.63 \times 10^{-34} \text{ J.s, } \lambda = 2 \times 10^{-10} \text{ m}$$

For an electron,

$$\text{momentum, } p = \frac{h}{\lambda} = \frac{6.63 \times 10^{-34}}{2 \times 10^{-10}} = 3.315 \times 10^{-24} \frac{\text{kg.m}}{\text{s}}$$

$$\text{Energy, } E = \frac{p^2}{2m} = \frac{(3.315 \times 10^{-24})^2}{2 \times 9.1 \times 10^{-31}} = 6.038 \times 10^{-18} \text{ J}$$

For an photon,

$$\text{momentum, } p = \frac{h}{\lambda} = \frac{6.63 \times 10^{-34}}{2 \times 10^{-10}} = 3.315 \times 10^{-24} \frac{\text{kg.m}}{\text{s}}$$

$$\text{Energy, } E = h\nu = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{2 \times 10^{-10}} = 9.945 \times 10^{-16} \text{ J}$$